

Chapter 3

Passive Stabilization with Hysteresis Rods

This section expands on the author’s original contributions on sizing, testing and validating a passive stabilization solution that can help in mitigating a tumbling condition due to the use of drag sail modules. The solution involves the installation of thin cylindrical rods made out of soft ferromagnetic materials that exhibit magnetically hysteretic behavior. This addition introduces a rate-dependent, magnetically induced dissipative torque to detumble a satellite and continuously provide stabilizing damping.

3.1 Fundamentals of Magnetic Hysteresis

When a ferromagnetic material is exposed in a varying magnetic field, the magnetic domains within the material undergo a series of changes. The magnetic domains are volumetric regions within the material that possess a common, distinct magnetic dipole in both magnitude and direction [20]. Magnetic domains are analogous to the grains found within a solid material’s micro-structure. Any magnetic domain region, theoretically, can be as small as the Bohr magneton, the fundamental magnetic quantity. The changing external field can enlarge/shrink the domains by displacement of their boundaries, rotate the domains to align with the direction of the applied field (i.e. easy magnetic axis), or eliminate part of a domain wall, thus forcing the domain to join an adjacent one. Some of these processes are reversible and some others irreversible, but all of those mechanisms have something in common; they lead to energy losses in the form of heat. It is important to note that the ferromagnetic material does not react instantaneously to the externally applied varying magnetic field; it rather follows the change with a time delay, generating a magnetic flux though its volume that is different from the externally applied field. The time delay with which the material reacts to the changing magnetic field gives rise to the definition of *magnetic hysteresis*. Under these conditions, the material will exert a torque that will try to *resist the change*

in the magnetic flux due to the changing magnetic field. The magnitude of the generated torque depends on the frequency and amplitude of the externally applied field, as well as the dimensions and other properties of the ferromagnetic material.

More specifically, a ferromagnetic material exhibiting hysteretic behavior is characterized by a sigmoid (i.e. arctangent) relationship between the external magnetic field strength and the induced magnetic flux density through the material's volume. This relationship is empirically summarized in the form of the characteristic *hysteresis major loop*, which is the region enclosed in the field-flux plot (H-B plot) bounded by this sigmoid relationship. An example of a major loop is illustrated in Figure 3.1 below. The horizontal axis of the H-B plot captures the value of the externally applied magnetic field strength, while the vertical axis captures the induced flux density within the soft ferromagnetic medium. It is noted that the term "soft" relates to the material's ability to *change* its dipole moment, *with a delay*, based on the direction and magnitude of the externally applied magnetic field. In contrast to permanent magnets (ferromagnetically "hard") whose dipole moment does not change expect when exposed to extreme temperatures or extreme magnetic fields, a soft hysteresis material will get magnetized and demagnetized based on the magnetic field that is exposed to. In the absence of a magnetic field, a soft ferromagnetic material will eventually demagnetize, which is not true for hard ferromagnetic materials that retain their dipole moment (i.e. they are more *coercive*).

The material's hysteretic behavior is bounded to reside within the limits of the characteristic major loop. The two outer sigmoid curves that form the major loop are known as the *right* (ascending) and *left* (descending) boundaries. Knowledge of the closed-form expression of the major loop is crucial for phenomenological modelling. It is important to note that different parts of the sigmoid curve correspond to and are dominated by a different magnetization process. For more details about the different magnetization stages of the hysteresis loop, consult [21], and [22]. For highly accurate modelling of hysteresis, each magnetization process is modelled separately, and the total magnetization is computed as the sum of all contributions from each process. The latter is outside the scope of this thesis.

Flattley and Henretty [23] have empirically demonstrated that the major hysteresis loop of a material is well approximated by an arctangent function. This arctangent model of ferromagnetic hysteresis is used for the simulation and analysis in subsequent sections.

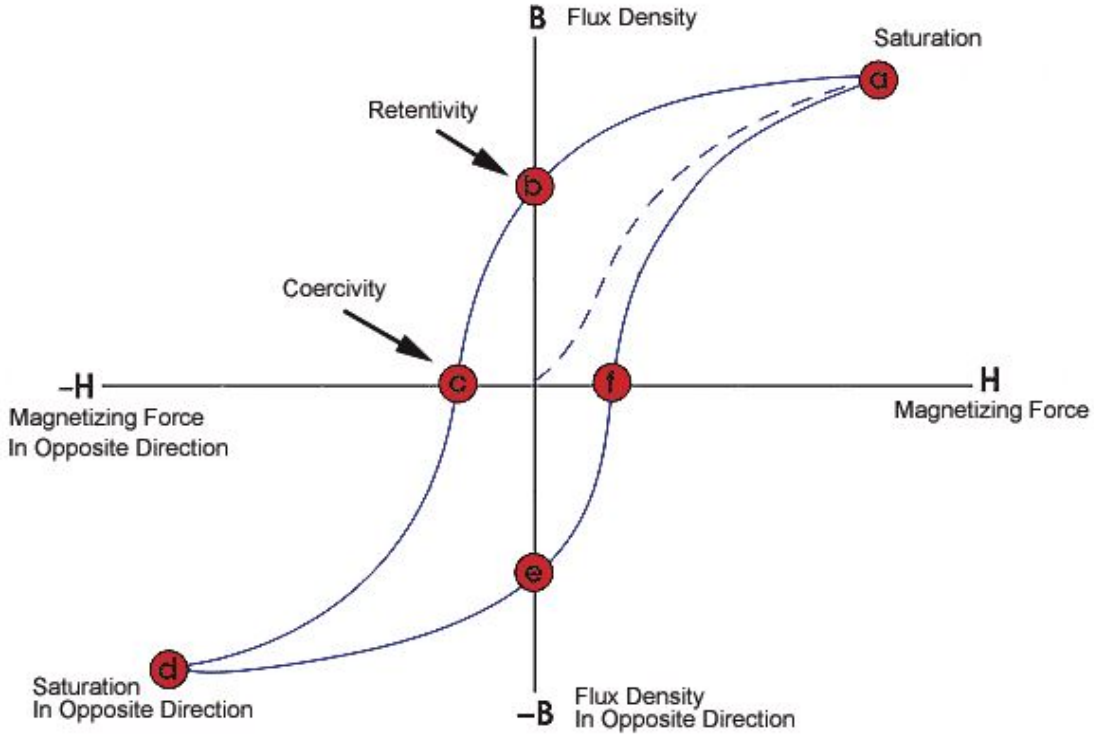


Figure 3.1: Example of a typical hysteresis major loop, characteristic of a soft ferromagnetic material [24]

3.2 Literature Review on Spacecraft Stabilization with Magnetic Hysteresis

The first satellites equipped with rods of soft ferromagnetic material to passively detumble was TRANSIT-1A and TRANSIT-1B [25]. In recent years, hysteresis rods have been used in the design of passive magnetic attitude control systems for CubeSats. A list of satellites that have used hysteresis rods as part of their attitude control system is provided in [21].

The empirical model of [23] is formulated by comparing and fitting simulated data with experimental data collected from a specific rod and test setup configuration. Kumar, Mezanej and Heck [25] later use the same model for the development of an attitude dynamics simulator. Santoni and Zelli [26] have experimentally determined the magnetic properties of hysteresis rods made of a specific nickel-iron alloy (*HyMu 80®*), presenting a methodology for modeling some of the hysteresis parameters of a rod based on its physical specifications, length and radius, rather than ideal material properties. Park, Seagraves and McClamroch[27] provide analytical expressions and the equations of motion for a satellite equipped with hysteresis rods as well as a permanent magnet, based on the work of [23] and [25] above.

With regards to testing the hysteresis rod efficiency, Santoni and Zelli [26], [28], Ivanov [29] and Gerhardt [30] all present different experimental setups. Of particular importance is the setup of the latter, making use of a Helmholtz cage to provide isolation, while also testing the effect of *mutual magnetization* when more than one rod is installed along a spacecraft axis.

With respect to the theoretical framework of hysteresis and domain theory, a qualitative approach is given by [31], while [21] and [22] also provide insight to the fundamental mechanisms that drive hysteretic behavior.

Finally, considering the post-launch performance verification of systems equipped with hysteresis rods, there are noticeable gaps reported between the expected and the actual passive stabilization results on-orbit. The Delfi-C3 cubesat required 60 days to achieve passive stabilization, while the simulated results were predicting detumbling in 1 orbit [32]. RAX-2 also witnesses a discrepancy between the expected and the actual stabilization time that was required for detumbling [33]. Much of these discrepancies seem to stem from poor characterization of the hysteresis rods, as well as magnetization of the rods due to the presence of a permanent magnet within the same satellite bus, mutual magnetization between multiple hysteresis rods, and the presence of other unmodelled sources of parasitic magnetism.

3.3 Simplification and Assumptions

The following study assumes that the hysteresis rods are mounted parallel to each of the spacecraft's orthogonal axes. This simplifies the rotation algebra required by eliminating one additional transformation from the spacecraft's body frame to the hysteresis rod's frame. This way, the latter frame does not need to be instantiated.

With regards to the use of multiple hysteresis rods per satellite body axis, the issue of mutual magnetization between the rods arises. Gerhardt and Palo [34] show experimentally that mutual magnetization between rods is insignificant when rods are perpendicular to each other. However, the phenomenon is significant when the rods are mounted parallel to each other and at a distance of less than 30%-40% of their length. This leads to decreased overall dipole efficiency (dipole moment per volume of hysteresis material). Considering that each rod needs to have the longest allowable possible length (roughly 150mm for NEMO class spacecraft) for maximum dipole efficiency, additional space limitations arise to fit the rods inside the spacecraft bus, as well as mass budget limitations. For this study, it is assumed that one rod per spacecraft body axis is used, for a total number of three rods mounted orthogonally within the spacecraft. It is shown that this simple design is sufficient to meet the detumbling requirement. It is noted that the requirement for this analysis is the cancellation and reduction of angular velocity of a satellite with the configuration provided at the end

of Chapter 2. No quantitative requirement has been established for the rate of detumbling due to the addition of hysteresis rods. However, a reasonable quantitative requirement is the goal to cancel out the spin-up behavior of a satellite equipped with drag sails, as explained in Chapter 2. Specifically for the spin-up study of NorSat-3, this requirement would be to size the hysteresis rods such that the spin-up conditions of linearly increasing body rates of 1 *deg/sec* per day are effectively nullified to zero body rates. Within this chapter, it is shown that this goal is successfully met.

When studying the phenomenon of hysteresis in ferromagnetic materials, the frequency dependent losses due to Eddy currents should also be taken into consideration. When the magnetizing field is changing quickly (i.e. the external magnetic field is not *quasi-static*), it will change the shape of the hysteresis loop due to the opposing effect of the induced field in the material. The magnetic domains, during the process of aligning their dipole with the externally applied magnetic field, induce Eddy currents which oppose the change of the magnetic field in the material. This leads to reduced effectiveness of the applied external excitation, leaving Eddy current losses to dominate, as the frequency of the magnetizing field increases. The problem was first examined theoretically by Shockley, Williams and Kittel (Williams et al 1950; Cullity, 1972), while Bertotti (Bertotti, 1998) examines the problem with statistical principles in mind. An analytical calculation of the Eddy Current losses is found in [22]. The algorithm has been integrated in simulation packages such as the open-source *Modelica Standard Library* and its effect can be quantitatively evaluated. A simulation example in *Open Modelica* that illustrates the effect of the frequency on the shape of the hysteresis curve is shown in Figure 3.2 below. The figure illustrates the hysteresis response of an soft ferromagnetic inductor when exposed to a sinusoidal magnetic field, at three distinct frequencies of 0, 10 and 100 *Hz*. As the frequency of the sinusoidal field increases, Eddy current losses dominate, leading to a drastic change in the hysteretic response of the material. This behavior is very important because it provides us with the important insight that testing of soft ferromagnetic materials needs to be performed at very slowly alternating magnetic fields. The Earth's magnetic field, in combination with the relative orbital motion of a satellite around the Earth and the satellite's attitude leads to the satellite's hysteresis rods being exposed a very slowly changing magnetic field. Therefore, the test equipment that needs to be used to characterize the hysteresis rod properties needs to produce a very slowly changing magnetic field with a period that is of the same order of magnitude as the combined effects of frequency of rotation (i.e. and by extension angular velocity) of the satellite and its orbital motion.

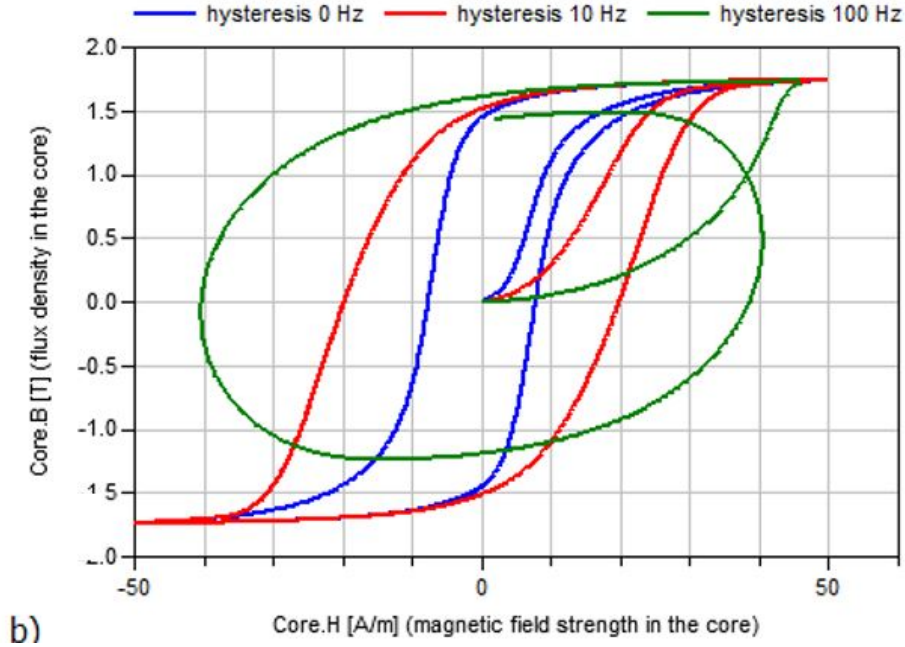


Figure 3.2: Simulated hysteresis curve of an inductor with a ferromagnetic core, for sinusoidal excitation fields of 0, 10 and 100 Hz [35]

With regards to NorSat-3, considering that the magnetic field does not change much in magnitude or direction locally in comparison to the angular velocity of the spacecraft, the frequency of the field can be loosely related to the spacecraft's body rate. For a single axis body rate of 10 deg/sec or 0.1745 rad/sec , the corresponding frequency of the field f is $f = \omega/2\pi = 0.023Hz$. The frequency is very low and therefore the modelling of Eddy current losses can be excluded from the simulation without any loss of fidelity. However, this points towards the need for reliable testing methodology to accurately determine the magnetization characteristics of the rods. Performing the test with high excitation frequencies would lead to the misleading result that the material is over-performing and that has the ability to dampen rotational motion faster than expected.

In subsequent sections, we review the equations of spacecraft attitude kinematics and dynamics. Then, we illustrate the equations of hysteresis damping torque. At first, we provide some necessary background on the notation used in the next sections, as well as the coordinate frames and relevant transformations from one frame to another.

3.4 Notation

Vectors are typically expressed in relation to a right-handed orthonormal set of vectors called a *frame*. In this thesis, *vetrix* notation adopted from Dr. Peter Hughes will be used to manipulate physical vectors in multiple reference frames [36]. A brief description of vetrices follows.

The *vectrix* $\underline{\mathcal{F}}_A$ corresponding to the frame $\mathcal{F}_A$ is given by:

$$\underline{\mathcal{F}}_A \triangleq \begin{bmatrix} \hat{\mathbf{a}}_1 & \hat{\mathbf{a}}_2 & \hat{\mathbf{a}}_3 \end{bmatrix}^T \quad (3.1)$$

A vectrix possesses simultaneously the characteristics of both a vector and a matrix. The *elements of the fundamental vectrix* $\underline{\mathcal{F}}_A$ are a set of *basis unit vectors*. Each of those orthonormal basis vectors is a column vector in three dimensional space. For instance, using the vectrix $\underline{\mathcal{F}}_A$, a three-dimensional vector $\underline{\mathbf{y}} \triangleq \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^T$ can be expressed in terms of its components in frame $\mathcal{F}_A$. For example, the vector $\underline{\mathbf{y}}$ is expressed in frame $\mathcal{F}_A$ with the components $\mathbf{v}_A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^T$ as:

$$\underline{\mathbf{y}} = \underline{\mathcal{F}}_A^T \mathbf{v}_A \quad (3.2)$$

Equation 3.2 holds for any frame of choice. Therefore, we can relate the components of $\mathbf{v}_A$ to the components of $\underline{\mathbf{y}}$ in some other frame $\mathcal{F}_C$, using the rotation matrix $\mathbf{C}_{AC}$ where $\mathbf{C}_{AC} = \underline{\mathcal{F}}_A \cdot \underline{\mathcal{F}}_C^T$ and $\mathbf{v}_A = \mathbf{C}_{AC} \mathbf{v}_C$. Rotation matrices provide one of the most fundamental representations of attitude; we can define a frame $\mathcal{F}_B$ fixed to a spacecraft body, and represent the attitude of the spacecraft in some arbitrary frame $\mathcal{F}_Q$ using $\mathbf{C}_{BQ}$. A rotation matrix can be parameterized by a set of angles called *Euler Angles*, θ_i , which describe a series of principal rotations (rotations about the frame axes). More specifically, the rotation matrix $\mathbf{C}_1(\theta_1)$ describes the rotation of a frame about the first axis (x -axis) of the frame by an angle equal to θ_1 . For the 3-2-1 *Yaw-Pitch-Roll* Euler rotation sequence, the rotation matrix $\mathbf{C}$ is given by:

$$\mathbf{C} = \mathbf{C}_1(\theta_1) \mathbf{C}_2(\theta_2) \mathbf{C}_3(\theta_3) \quad (3.3)$$

Another parameterization is the Euler axis angle, according to which the rotation of a body can be expressed as a rotation of ϕ about an axis along $\mathbf{a}$, according to:

$$\mathbf{C} = \cos \phi \mathbf{1} + (1 - \cos \phi) \mathbf{a} \mathbf{a}^T - \sin \phi \mathbf{a}^\times \quad (3.4)$$

An alternative attitude parametrization is the use of *quaternions*, which are used extensively for attitude simulations due to their computational efficiency, and lack of singularities for rotation sequences. In this case, a rotation sequence can be described using the 4-parameter set $\mathbf{q} = \begin{bmatrix} \boldsymbol{\epsilon}^T & \eta \end{bmatrix}^T$, where:

$$\boldsymbol{\epsilon} = \mathbf{a} \sin\left(\frac{\phi}{2}\right), \quad \eta = \cos\left(\frac{\phi}{2}\right) \quad (3.5)$$

A thorough treatment on the properties of frames, vectors, vectrices, and rotations can be found in [36].

Unless otherwise explicitly noted, bold and lower-case letters will be used to denote column matrices, bold and capital letters will be used to denote two-dimensional matrices. Finally, scalar values will be denoted as italicized, non-bold letters, with occasional subscripts where applicable.

3.5 Coordinate Frames

In this section, we describe the coordinate frames that are relevant to our analysis. Some frames are related to orbital propagation, while others are related to the spacecraft's attitude. Throughout this thesis, the term *coordinate frame* will be used interchangeably with the term *coordinate system*. A *coordinate system* is the realization of a coordinate frame centered at a specific origin point in space, using a triplet of orthonormal right-handed basis vectors. Left-handed coordinate systems exist as well, but they do not form part of this discussion, since they are not commonly encountered. For example, some manufacturers of rate sensor or magnetometer components use a left-handed coordinate system to define the positive direction of the units' measurements positive direction on each axis.

The discussed coordinate systems are distinguished in two categories; Earth-based systems and satellite-based systems. Earth-based systems are realized with their origin attached at the center of the Earth, while satellite-based systems are realized with their origin attached at the satellite's Center of Mass (CoM).

3.5.1 Earth-Centered Inertial (ECI) Frame $\mathcal{F}_i$

The *Earth-Centered Inertial* (ECI) system $\mathcal{F}_i$ is also known as *geocentric-equatorial* system [37]. The origin of the system is the center of the Earth, and it is defined by the Earth's axis of rotation and its X-axis pointing towards the vernal equinox. The Z-axis points towards the North pole, and the Y-axis completes the right-hand rule. Even though $\mathcal{F}_i$ is attached at the Earth's center, this system does not rotate with the Earth. A realization of the ECI frame can be quite intricate and laborious, because the fundamental X-Y plane and the vernal equinox move slowly over time. Therefore, any realization must specify the direction of the vernal equinox at a specific *epoch* (date), leading up to the realization of a quasi-inertial or *pseudo-inertial* frame, in reality. For example, the *J2000* frame is a popular quasi-inertial frame instantiated with the direction of the vernal equinox captured precisely at 12:00 January 1st 2000. Its motion is modeled using the IAU-1976 model (for modelling the precession of the Earth's axis of rotation) and the IAU-1980 model (for modelling the nutation of the Earth's axis of rotation). For more information about the effects of precession, nutation, and polar motion, the reader can consult [37]. The above perturbations also play a role in the coordinate frame transformation from ECI to any other frame (e.g. ECEF), and vice versa. In addition, different applications might allow for a number of simplifications in

terms of whether some of these effects are taken into account, according to the required level of detail.

3.5.2 Earth-Centered Earth-Fixed Frame $\mathcal{F}_e$

The *Earth-Centered-Earth-Fixed* (ECEF) frame $\mathcal{F}_e$ is also known as the *International Terrestrial Reference Frame* (ITRF). It is defined with its origin at the center of the Earth. Its X axis is aligned with the Prime Meridian, while its Z axis is aligned with the North Pole. Finally, the Y axis is instantiated by completing the right-hand rule. The ECEF frame rotates with respect to the ECI frame at an approximately constant angular velocity $\omega_\oplus$ about the Z axis. The angular velocity vector describing the motion of $\mathcal{F}_e$ with respect to $\mathcal{F}_i$, expressed in $\mathcal{F}_i$, is denoted as $\underline{\omega}_{ei}$ and is defined as:

$$\underline{\omega}_{ei} = \underline{\mathcal{F}}_e^T \begin{bmatrix} 0 & 0 & \omega_\oplus \end{bmatrix}^T \quad (3.6)$$

$$\omega_\oplus = 7.292115 \times 10^{-5} \text{ rad/s (all digits significant)} \quad (3.7)$$

3.5.3 Local Vertical Local Horizontal Frame $\mathcal{F}_o$

In contrast to the ECEF $\mathcal{F}_e$ and ECI $\mathcal{F}_i$ frames, the LVLH frame $\mathcal{F}_o$ is a satellite-based system. Its origin is centered at the satellite's center of mass, and its realization is easiest formulated by using the satellite's position and velocity vectors with respect to $\mathcal{F}_i$ and expressed in $\mathcal{F}_i$. The vectrix $\underline{\mathcal{F}}_o$ corresponding to the LVLH frame $\mathcal{F}_o$ is given by:

$$\underline{\mathcal{F}}_o = \begin{bmatrix} \hat{\mathbf{q}}_1 & \hat{\mathbf{q}}_2 & \hat{\mathbf{q}}_3 \end{bmatrix}^T \quad (3.8)$$

The basis unit vector $\hat{\mathbf{q}}_3$, which is the frame's Z-axis, is aligned with the spacecraft geocentric position vector, but pointing towards the Earth (opposite direction). The basis vector $\hat{\mathbf{q}}_2$ is aligned with the negative direction of the orbit normal. The orbit normal vector is the cross product of the position and velocity vectors expressed in the ECI frame. Finally, the basis vector $\hat{\mathbf{q}}_1$, the frame's X-axis, completes the right hand rule with respect to the other two basis vectors.

3.5.4 Spacecraft Body Frame $\mathcal{F}_b$

The satellite's body frame is denoted as $\mathcal{F}_b$. The origin of the body frame is attached to its CoM. The basis vectors of $\mathcal{F}_b$ are aligned with the orthogonal faces of the cuboid rectangular satellite bus structure. The vectrix $\underline{\mathcal{F}}_b$ corresponding to the body frame $\mathcal{F}_b$ is given by:

$$\underline{\mathcal{F}}_b = \begin{bmatrix} \hat{\mathbf{b}}_1 & \hat{\mathbf{b}}_2 & \hat{\mathbf{b}}_3 \end{bmatrix}^T \quad (3.9)$$

$\hat{\mathbf{b}}_1$ is aligned with the spacecraft X axis, $\hat{\mathbf{b}}_2$ is aligned with the spacecraft Y axis, and $\hat{\mathbf{b}}_3$ is aligned with the satellite's Z axis. It is noted that the body frame axes do not generally align with the body's principal axes, the orthogonal triplet of basis vectors for which a diagonal moment of inertia matrix can be defined (i.e. the matrix composed by the eigen-vectors of the satellite's inertia tensor). Many times, payload and mission requirements as well as other mechanical constraints (e.g. the deployment mechanism) restrict the layout options for the placement of the satellite's components. This leads inevitably to misalignment between the body frame and the principal axes frame. As we will examine in Chapter 4, mechanical layout constraints, lack of space, and bulky payload instruments can drive design choices that may lead to death modes, requiring thorough evaluation and mitigation strategies.

3.6 Coordinate Transformations

Having defined the coordinate systems, this section describes the equations needed to manipulate and express quantities between those frames.

3.6.1 ECEF to ECI Frame, $\mathcal{F}_e$ to $\mathcal{F}_i$

In the context of simulating spacecraft attitude perturbations, the conversion of quantities from the ECEF frame $\mathcal{F}_e$ to the ECI frame $\mathcal{F}_i$ is important. For example, when computing the magnetic field flux density of the Earth during satellite propagation, this value is first calculated and expressed in the $\mathcal{F}_e$ frame using models of the Earth's geomagnetic potential. However, the equations governing the satellite's orbital motion are expressed in the $\mathcal{F}_i$ frame. The conversion from $\mathcal{F}_e$ to $\mathcal{F}_i$ (specifically the J2000 frame) and vice-versa is a delicate procedure that takes into account a series of phenomena that are described by a specific sequence of coordinate transformations. First of all, the sidereal rotation of the Earth is taken into account, expressed as the coordinate transformation matrix $\mathbf{C}_\Theta$. In addition, Earth axis nutation transformation is expressed as $\mathbf{C}_\Phi$, while the Earth's precession is expressed as $\mathbf{C}_N$. Finally, the effect of the Earth's polar motion is expressed as $\mathbf{C}_\Omega$. The rotation sequence of those transformations is fundamentally important. For a satellite's position vector $\mathbf{r}_e$ expressed in the ECEF frame $\mathcal{F}_e$, the coordinate transformation $\mathbf{C}_{ie}$ from $\mathcal{F}_e$ to $\mathcal{F}_i$ to get the position vector $\mathbf{r}_i(t)$ in $\mathcal{F}_i$ is expressed as:

$$\mathbf{r}_i(t) = \mathbf{C}_{ie}(t)\mathbf{r}_e = \mathbf{C}_\Phi(t)\mathbf{C}_N(t)\mathbf{C}_\Theta(t)\mathbf{C}_\Omega(t)\mathbf{r}_e \quad (3.10)$$

The above process is referred in the literature as the *FK5 reduction*. It can also be encountered as *IAU-76/FK5 reduction*, due to the use of the *IAU-76* model to describe the Earth's precession, nutation, and polar motion. For a detailed treatment of those transformations,

as well as a discussion on the various models that the analyst can choose to realize those transformations, consult [37]. There is general consensus that, especially for the modelling of the polar motion, the models are not very precise for long propagation intervals into the future. The associated coefficients, models, and other constants that are needed for realizing these transformations need to be frequently updated. Typically, the latter data are provided at regular intervals from the International Earth Rotation and Reference systems Service, also abbreviated as *IERS* [38]. Depending on the required application and level of detail, analysts should be aware about the effect of data updates to the validity of results.

3.6.2 ECI to LVLH Frame, $\mathcal{F}_i$ to $\mathcal{F}_o$

As discussed in Section 3.5.3, the LVLH frame is easily realized from the ECI frame, therefore the transformation is straightforward. Consider the satellite's position and velocity vectors $\mathbf{r}_i$ and $\mathbf{v}_i$, expressed in the ECI frame $\mathcal{F}_i$. The angular momentum $\mathbf{h}_i$ of the satellite expressed in the ECI frame *Hfi* is defined as the cross product of the position and velocity vectors, according to $\mathbf{h}_i = \mathbf{r}_i \times \mathbf{v}_i$. The representation of the LVLH basis vectors with respect to $\mathbf{r}_i$, $\mathbf{v}_i$ and $\mathbf{h}_i$ is given by:

$$\mathbf{o}_{3,i} = -\frac{\mathbf{r}_i}{\|\mathbf{r}_i\|}, \quad \mathbf{o}_{2,i} = -\frac{\mathbf{h}_i}{\|\mathbf{h}_i\|}, \quad \mathbf{o}_{1,i} = \mathbf{o}_{2,i} \times \mathbf{o}_{3,i} \quad (3.11)$$

The rotation matrix from the ECI frame $\mathcal{F}_i$ to the LVLH frame $\mathcal{F}_o$ then becomes:

$$\mathbf{C}_{oi} = \begin{bmatrix} \mathbf{o}_{1,i} & \mathbf{o}_{2,i} & \mathbf{o}_{3,i} \end{bmatrix}^T \quad (3.12)$$

3.6.3 ECI to Body Frame, $\mathcal{F}_i$ to $\mathcal{F}_b$

The rotation from the inertial frame $\mathcal{F}_i$ to the spacecraft body frame $\mathcal{F}_b$ is written as $\mathbf{C}_{bi}$ and it is a fundamental expression of spacecraft attitude. Its computation forms part of the attitude estimation algorithms running on the spacecraft's on-board computer. More insight on the operation of OASYS and the on-board computation of the attitude determination algorithms is provided in [39].

3.7 Spacecraft Orbital Dynamics

This section provides a brief discussion of fundamental orbital mechanics concepts. For a detailed treatment, consult [37] and [40]. In addition, we provide a high-level description about SGP4, the closed-form orbital propagation algorithm that is used for all simulations of this thesis.

3.7.1 Orbital Elements and Equations of Motion

The starting point for orbital dynamics is the equation describing Keplerian two-body motion. Keplerian orbital motion is planar motion along an elliptical track, with the Earth placed at one of the ellipse's foci. It is derived from classical Newtonian mechanics, starting with the motion of a point mass about an orbiting body. The spacecraft's orbit can be expressed using six orbital elements, also known as Keplerian elements. The sixth element, called the *mean anomaly*, describes the time-varying component of the spacecraft's orbital position. Figure 3.3 visualizes three of the seven orbital elements, namely the inclination i , the right ascension of the ascending node Ω , and the argument of perigee ω . These three angles describe the orientation of the orbit plane with respect to the primary body. The inertial frame $\mathcal{F}_i$ is also shown in the image for reference. Finally, the orbit's angular momentum vector $\underline{\mathbf{h}}_i$ can be shown, as it was defined in Section 3.6.2 above.

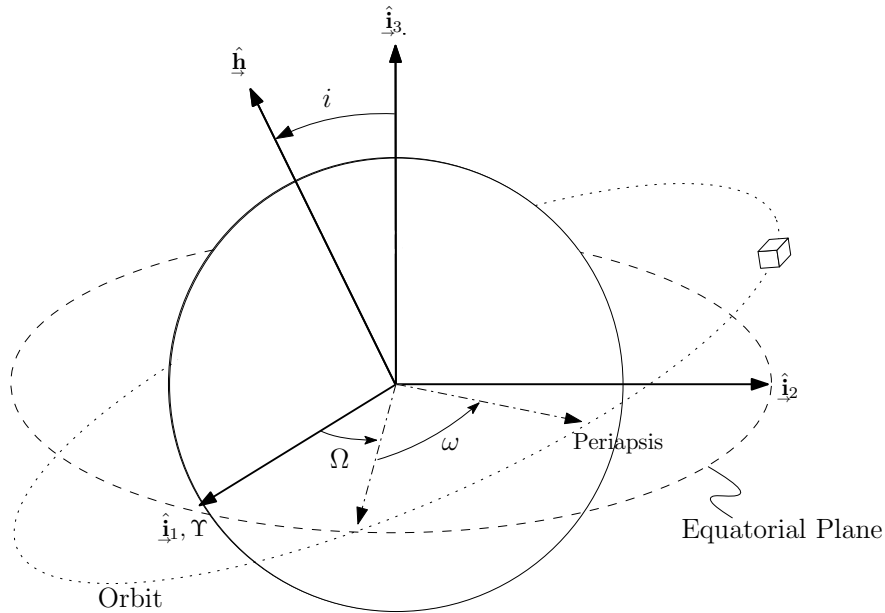


Figure 3.3: Depiction of orbital elements, i , Ω , ω [39]

A brief description of the orbital elements follows:

- a - The *semi-major axis* of the orbit's ellipse. The semi-major axis is the parameter that specifies the *size* of the orbit.
- e - *eccentricity*: a measure of a conic section's deviation from being circular, $0 < e < 1$ for an ellipse [40].
- i - *inclination*: the angle between the equatorial plane and the orbital plane, or equivalently the angle between the basis vector $\hat{\mathbf{i}}_3$ of the ECI frame $\mathcal{F}_i$, and $\underline{\mathbf{h}}$

- Ω - *Right Ascension of the Ascending Node (RAAN)*: the angle in between the vernal equinox, $\hat{\mathbf{i}}_1$ and the vector $\hat{\mathbf{n}}$, which points from the Earth's center of mass towards the ascending node (the point where the orbit crosses the equatorial plane, and the coordinate along $\hat{\mathbf{i}}_3$ crosses from negative to positive)
- ω - *argument of perigee*: the angle between the ascending node and the perigee
- M_0 - *initial mean anomaly*: the initial mean anomaly $M_0 = M(t_0)$, which is typically used in place of the true anomaly. The *true anomaly* is the angle between perigee and the position of the orbiting body, at any given time.

3.7.2 Keplerian Two-body Equation of Motion

In order to propagate the satellite's orbital state as a Keplerian two-body motion, the following differential equation needs to be solved [40]:

$$\ddot{\mathbf{r}} = -\frac{\mu}{\|\mathbf{r}\|^3}\mathbf{r} \quad (3.13)$$

Where μ is the gravitational constant of the primary body. [40] provides the analytical solution to the above equation.

The Keplerian two-body propagator is sufficiently accurate for short propagation time intervals up to several days, but it quickly becomes inaccurate due to the absence of adequate modelling for certain perturbations. In reality, a satellite's orbit is affected by other factors that make the orbit deviate from the ideal Keplerian motion. These factors include the non-spherical shape of the Earth, the presence of other celestial bodies and their gravitational fields, atmospheric drag, etc. Accurate modelling of orbit perturbations within the orbital propagator requires detailed numerical solutions to non-linear differential equations, whose computation time scale with complexity.

3.7.3 SGP4 Propagation Theory

As an alternative to the numerical propagation algorithms needed for highly accurate propagation, there exist closed-form analytical solutions that are sufficiently accurate for some applications. Since the 1970s, the US Air Force has been using the Simplified General Perturbation Theory to propagate a satellite's state using a set of initial orbital conditions. This set is packaged in a special, compact format and is known as Two Line Element (TLE) set. The official US SGP4 algorithm is open-source and is meant to be used as a standard for interoperability among satellite operators and analysts.

The SGP4 propagator is modelling the effect of the non-sphericity of the Earth up to a certain degree, and it also models atmospheric drag. For more information regarding the SGP4 Theory, consult [41]. The simulations performed in this thesis make use of the SGP4

algorithm as published and updated in the above reference. For more details about the definition and use of a Two Line Element, consult [41].

It is important to note that the accuracy of SGP4 drops over time, with an approximate error of 5 kilometers per day of propagation [41]. This is an acceptable level of accuracy for the purpose of our simulations, which span a maximum of 5 days. The accumulated error can be as high as 25 kilometers, which is acceptable since the key metrics that are evaluated in our study relate mostly to the spacecraft attitude dynamics.

3.8 Spacecraft Attitude Dynamics

This section describes briefly the equations of motion that describe the attitude of the spacecraft, treated as a rigid body.

First of all, the kinematic equation for the rotation from ECI frame $\mathcal{F}_i$ to Body Frame $\mathcal{F}_b$ is given by:

$$\dot{\mathbf{C}}_{bi} + \boldsymbol{\omega}^\times \mathbf{C}_{bi} = \mathbf{0} \quad (3.14)$$

Where $\boldsymbol{\omega}$ is the angular velocity of the spacecraft with respect to the ECI frame $\mathcal{F}_i$ and expressed in the body frame $\mathcal{F}_b$.

We define $\boldsymbol{\tau}_\bullet$ as the net external torque acting about the satellite's center of mass. We also denote $\mathbf{I}$ as the inertial matrix of the satellite expressed in the body frame $\mathcal{F}_b$. The equation describing the attitude dynamics of the spacecraft in relation to the net torque applied by all disturbance torques is called the *Euler's rigid body equation* and is defined as:

$$\boldsymbol{\tau}_\bullet = \mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}^\times \mathbf{I}\boldsymbol{\omega} \quad (3.15)$$

For the purpose of this analysis, no reaction wheel dynamics have been incorporated in the equations of motion. This is because we are interested in analyzing the spacecraft attitude without any means of active control mechanisms. For a formal treatment on the equations of motion using reaction wheel dynamics, consult [36], [39] and [42].

3.9 Environmental Disturbance Torques

The space environment imposes external torques that affect the attitude of a spacecraft. The key disturbance torques to consider are those caused by the Earth's magnetic field, aerodynamic effects, the gravity-gradient, and solar radiation pressure. For a complete discussion on attitude disturbance torques, please consult [39]. In this section, only two torques will be discussed. Simulations using SFL's MIRAGE simulation environment consider all disturbance torques to accurately model and control the spacecraft attitude.

3.9.1 Magnetic Field Torque

A magnetic disturbance torque can arise from the interaction of the Earth's magnetic field $\mathbf{b}$ with the magnetic dipole moment of the spacecraft. Magnetic moments can occur from electrical current loops in the hardware wiring (e.g. solar panels), as well as the presence of ferromagnetic materials. The magnetic disturbance torque $\underline{\tau}_m$ is calculated from the cross product of the spacecraft magnetic dipole moment $\underline{\mathbf{m}}_{net}$ and the local magnetic field.

$$\underline{\tau}_m = \underline{\mathbf{m}}_{net} \times \mathbf{b} \quad (3.16)$$

Permanent magnets (hard ferromagnetic materials) installed on the satellite contribute to $\underline{\mathbf{m}}_m$ and so do current loops. Some spacecraft equipment can produce magnetic moments at low levels during their operation, such as parasitic magnetic dipole moment stemming from the operation of reaction wheels or other components for radio communication with the ground station.

3.9.2 Hysteresis Damping Torque

This section describes the equations for the computation of the hysteresis torque. This torque is computed separately from the magnetic field torque, due to the nature of the dipole moment. While the magnetic field torque aggregates the effects of parasitic dipole moment and the presence of hard ferromagnetic materials (permanent magnets), the hysteresis damping torque aggregates the produced dipole moments only due to soft ferromagnetic materials.

Recall that we have assumed that the hysteresis rods are aligned with the spacecraft's body frame $\mathcal{F}_b$. This is practical in terms of mechanical layout and mounting of the rods on the satellite bus, requiring simple design of mounting brackets. When a spacecraft equipped with ferromagnetic hysteresis rods is immersed in a magnetic field, the hysteresis torque produced is identified as $\underline{\tau}_{hyst}$. Additionally, we define the magnetic flux density produced within the ferromagnetic material as $\mathbf{b}$. Then, the hysteresis torque can be computed as:

$$\underline{\tau}_{hyst} = \underline{\mathbf{m}}_{hyst} \times \mathbf{b} \quad (3.17)$$

In matrix notation, Equation 3.17 becomes:

$$\tau_{hyst} = \mathbf{m}_{hyst}^\times \mathbf{b} \quad (3.18)$$

τ_{hyst} is a 3×1 column matrix with each component defining the hysteresis torque that is produced about the corresponding spacecraft body axis and expressed in the body frame $\mathcal{F}_b$. $\mathbf{m}_{hyst}^\times$ is the skew symmetric matrix of $\mathbf{m}_{hyst}$, a 3×1 column matrix with each of its components representing the dipole moment that is produced due to the presence of the hysteresis rods. In addition, $\mathbf{b}$ is a column matrix defining the vector of the Earth's magnetic

field flux density in vacuum. All quantities in Equations 3.17 and 3.18 are expressed in the satellite's body frame $\mathcal{F}_b$. The SI units of $\mathbf{b}$ are in $N/(A \cdot m)$, also known as *Tesla* and abbreviated as *T*. The SI units of the magnetic moment $\mathbf{m}_{\text{hyst}}$ are in $A \cdot m^2$. The units of torque are in $N \cdot m$.

For reference, the typical symbol for magnetic flux density is B , in the field of electromagnetism. Here, abiding to Dr. Peter Hughes notation, we will be using $\mathbf{b}$ to make it clear that it is a column matrix. In vacuum, the magnetic flux density $\mathbf{b}$ is related to the magnetic field strength $\mathbf{h}$ through the magnetic permeability of vacuum constant μ_0 , as:

$$\mathbf{b} = \mu_0 \mathbf{h} \quad (3.19)$$

$$\mu_0 = 4\pi \times 10^{-5} \frac{N}{A^2} \quad (3.20)$$

The SI units of the magnetic field strength $\mathbf{h}$ are in A/m . The SI units of the μ_0 constant are N/A^2 . For reference, the typical symbol for magnetic field strength is H in the field of electromagnetism, but as discussed, we will be abiding to Dr. Peter Hughes notation and will be using the symbol $\mathbf{h}$ to make it clear that it is a column matrix.

It is reminded that the magnetic dipole moment due to the hysteresis rods $\mathbf{m}_{\text{hyst}}$ is a 3×1 column matrix that sums all contributions of dipole moment from all hysteresis rods mounted on the spacecraft structure, expressed in $\mathcal{F}_b$. At this point, we will explain how the dipole moment is computed for a single rod, before we move on to a formulation involving the installation of multiple rods.

When an external magnetic field $\mathbf{h}$ is applied to a hysteresis rod of volume v , the flux density produced within the rod is defined as $\mathbf{b}_{\text{rod}}$. Then, the magnetic dipole moment $\mathbf{m}_{\text{rod}}$ produced by the hysteresis rod is given by:

$$\mathbf{m}_{\text{rod}} = \frac{1}{\mu_0} v \mathbf{b}_{\text{rod}} \quad (3.21)$$

An important note needs to be made with respect to the properties of soft ferromagnetic materials. As discussed in the beginning of this chapter and by observing Figure 3.1, there are three important scalar values that characterize any soft ferromagnetic material; *saturation*, *coercivity*, and *remanence*. *Saturation* refers to the maximum amount of induced flux density that a soft ferromagnetic material can experience within any sufficiently large externally applied magnetic field. Saturation is denoted by b_s throughout the scope of this thesis. The points of saturation can be visualized as the points "a" and "d" in Figure 3.1. *Coercivity* is defined as the maximum amount of magnetic field strength for which the soft ferromagnetic material experiences zero magnetic flux density, when exposed to a periodically varying magnetic field. The points of coercivity can be visualized as the points "c" and "f" in Figure 3.1. Coercivity is denoted as h_c within this thesis. *Remanence*, or *Retentivity* is the

amount of maximum residual magnetic flux in the soft ferromagnetic medium, even when the externally applied field is of zero magnetic field strength. Remanence is denoted as b_r within this thesis, and can be visualized as the points "b" and "b" in Figure 3.1. Using these three parameters, a simple empirical arctangent model of hysteresis magnetization can be created. This empirical model, as described in [23] and [25] forms the basis of modelling the induced flux density of soft ferromagnetic materials in this simulation.

Note that in equation 3.21, we assume that the orientation of the rod is aligned with the orientation of the externally applied field. Therefore, we need to proceed and generalize the computation of the dipole moment for any number of rods installed, as well as their dimensions and orientation within the satellite body frame $\mathcal{F}_b$.

We define $\mathbf{v}$ as a $n \times 1$ column matrix with components $\mathbf{v} = \begin{bmatrix} v_1 & v_2 & \dots & v_{n-1} & v_n \end{bmatrix}^T$, where n is the number of hysteresis rods that have been installed on the satellite. Each component of the column matrix $\mathbf{v}$ represents the volume of each cylindrical hysteresis rod. For the purpose of this thesis, all rods are cylindrical in shape. However, it is possible that each rod may be different in terms of shape or dimensions, for example, in cases where there would be an interest to provide stronger hysteresis damping along one of the spacecraft axes. The formulation below holds without loss of generality.

Each of the n rods installed on the satellite has a specific orientation of its cylindrical longitudinal axis with respect to the satellite body frame $\mathcal{F}_b$. In some literature resources like [26], the longitudinal axis of an elongated soft ferromagnetic specimen is also encountered as the *easy magnetization axis*.

We define the matrix $\mathbf{W}$ as the *rod orientation matrix*, with dimensions $n \times 3$ and components:

$$\mathbf{W} = \begin{bmatrix} \hat{\mathbf{w}}_1 & \hat{\mathbf{w}}_2 & \dots & \hat{\mathbf{w}}_{n-1} & \hat{\mathbf{w}}_n \end{bmatrix}^T \quad (3.22)$$

For example, for the j^{th} hysteresis rod, $\hat{\mathbf{w}}_j$ is a column unit vector with components $\hat{\mathbf{w}}_j = \begin{bmatrix} \hat{\mathbf{w}}_{j,1} & \hat{\mathbf{w}}_{j,2} & \hat{\mathbf{w}}_{j,3} \end{bmatrix}^T$ describing the orientation of the j^{th} rod's longitudinal axis with respect to the spacecraft body frame $\mathcal{F}_b$. It is clear that all of the hysteresis rods will be exposed to the same magnetic field strength vector $\mathbf{h}$, but their response and resulting hysteresis moment will differ depending on the dimensions and their orientation, as hinted by Equation 3.21. They may also differ in terms of material properties, namely having different values of saturation b_s , coercivity h_c and remanence b_r .

At this point, an important simplification needs to be made. If we assume that all the rods are made out of the same material, then all of the three parameters of magnetic remanence b_r , coercivity h_c and saturation b_s can be the same for all rods. Under this assumption, we only need to compute a single vectorial value for the induced flux density, and then distribute the effect of that induced flux density according to the rod orientations and dimensions, as

shown below.

We define $\mathbf{b}_h$ as the induced flux density due to the Earth's magnetic field within the soft ferromagnetic material that is considered. It is a column vector with components $\mathbf{b}_h = \begin{bmatrix} b_{h,1} & b_{h,2} & b_{h,3} \end{bmatrix}^T$, expressed in the spacecraft body frame $\mathcal{F}_b$. Note that $\mathbf{b}_h$ has the same direction as the Earth's magnetic field $\mathbf{h}$.

At this point, we define $\mathbf{B}$ as the *flux density distribution matrix*, with dimensions $n \times 3$ and components:

$$\mathbf{B} = \begin{bmatrix} \mathbf{b}_{\text{in},1} & \mathbf{b}_{\text{in},2} & \dots & \mathbf{b}_{\text{in},n-1} & \mathbf{b}_{\text{in},n} \end{bmatrix}^T \quad (3.23)$$

For the j^{th} hysteresis rod, $\mathbf{b}_{\text{in},j}$ is a column vector with components $\mathbf{b}_{\text{in},j} = \begin{bmatrix} \mathbf{b}_{j,1} & \mathbf{b}_{j,2} & \mathbf{b}_{j,3} \end{bmatrix}^T$ describing the induced flux density of the j^{th} rod's along its longitudinal axis with respect to the spacecraft body frame $\mathcal{F}_b$ axes. According to the above, $\mathbf{B}$ can then be computed as:

$$\mathbf{B} = \mathbf{W} \text{diag}(\mathbf{b}_h) = \begin{bmatrix} \hat{\mathbf{w}}_1 & \hat{\mathbf{w}}_2 & \dots & \hat{\mathbf{w}}_{n-1} & \hat{\mathbf{w}}_n \end{bmatrix}^T \begin{bmatrix} b_{h,1} & 0 & 0 \\ 0 & b_{h,2} & 0 \\ 0 & 0 & b_{h,3} \end{bmatrix} \quad (3.24)$$

With the above formulation, $\mathbf{B}$ encapsulates all vectors of induced magnetic flux contributed by each hysteresis rod, expressed in the body frame $\mathcal{F}_b$. In the end, the magnetic moment $\mathbf{m}_{\text{hyst}}$ due to all hysteresis rods can be summarized as:

$$\mathbf{m}_{\text{hyst}} = \frac{1}{\mu_0} \mathbf{v}^T \mathbf{B} \quad (3.25)$$

Using Equations 3.19, 3.24 and 3.25 into Equation 3.18, we get:

$$\boldsymbol{\tau}_{\text{hyst}} = \left(\frac{1}{\mu_0} \mathbf{v}^T \mathbf{B} \right)^\times \mu_0 \mathbf{h} = \left(\mathbf{v}^T \mathbf{W} \text{diag}(\mathbf{b}_{\text{in}}) \right)^\times \mathbf{h} \quad (3.26)$$

For details on the computation of the induced flux density vector $\mathbf{b}_{\text{in}}$ consult [23], [25], and [27]. The process involves the solution to a non-linear differential equation based on the arctangent model of the hysteresis major loop, using the values of saturation b_s , coercivity h_c , and remanence b_r .

3.10 Hysteresis Rod Sizing

By observation of the hysteresis torque Equation 3.26, we note that the magnitude of the hysteresis torque is linearly dependent on the volume of soft ferromagnetic material used. The equation, however, does not clearly capture the equally important concept of *dipole efficiency* and the effect to which the law of diminishing returns is in effect. With regards

to cylindrical ferromagnetic rods (as well as thin ferromagnetic strips), experiments in [26] and [28] have demonstrated that *long and thin specimens produce higher dipole moment per volume*. This leads to a need to conduct experiments and assess the performance of an array of small size hysteresis specimens [28] versus the use of a single larger specimen. Arguments defending the use of single rods are based on the reduced complexity during spacecraft integration, as well as the negative and difficult to quantify effect of mutual magnetization between rods/strips mounted in parallel. Experiments conducted by Gerhardt [30] in a Helmholtz cage show that mutual magnetization can reduce dipole efficiency of each rod if the rods are installed at a distance of less than 30-40% of each specimen's length.

Specifically for NorSat-3, the initially considered rod length is based on hysteresis rods considered in trade studies of the EV9 mission, using the GNB bus (200 x 200 x 200 mm). The NorSat-3 bus nominal dimensions are 200 x 300 x 400 mm, therefore, if the rods are intended to be the same length, they all need to conform to the shortest bus dimension, leading back to the length chosen for EV9. The length of each rod planned to be used is approximately 152 mm, accounting for the thickness of the bus structure. If multiple rods were considered in order to double the dipole moment (and consequently the mass of the passive solution), then each rod would have to be spaced about 53mm apart. For using two rods, for example, each assembly would require a clear area of 152 x 53 mm for their installation on a panel with no other mechanical interferences. If three rods would be used, then this area would be 152 x 106 mm. Bracket size and mass would also increase with increasing number of rods. Available space and mass in the bus is always at a premium in microsatellite applications. Therefore, a single rod per satellite body axis is assumed to be used in the simulations, leaving the rod diameter as the last free variable in the trade space for optimal dipole moment efficiency. The simulation results prove that a single rod per axis is sufficient in achieving spin-up cancellation.

Under those assumptions, the rod diameter will determine both the rod dipole efficiency and the dipole moment, which are two somewhat conflicting figures of merit. Due to the availability of preexisting inventory of hysteresis rods at SFL, as well as the availability of a test setup with sense coils of the exact dimensions, specimens of 6.4 mm in diameter are used. The resulting length-to-diameter ratio l/D for a rod of length $l = 6.4mm$ and diameter $D = 152mm$ is therefore:

$$\frac{l}{D} = \frac{152 \text{ mm}}{6.4 \text{ mm}} = 23.75 \simeq 24 \quad (3.27)$$

According on [26], a length-to-diameter ratio of 24 is considered fairly low and will lead to some loss of dipole efficiency (dipole moment per unit of volume) in comparison to a rod that would be thinner. Of course, a thinner rod would produce less dipole moment, and considerations of multiple rods, mass budget discipline, volume space savings and mutual magnetization arise.

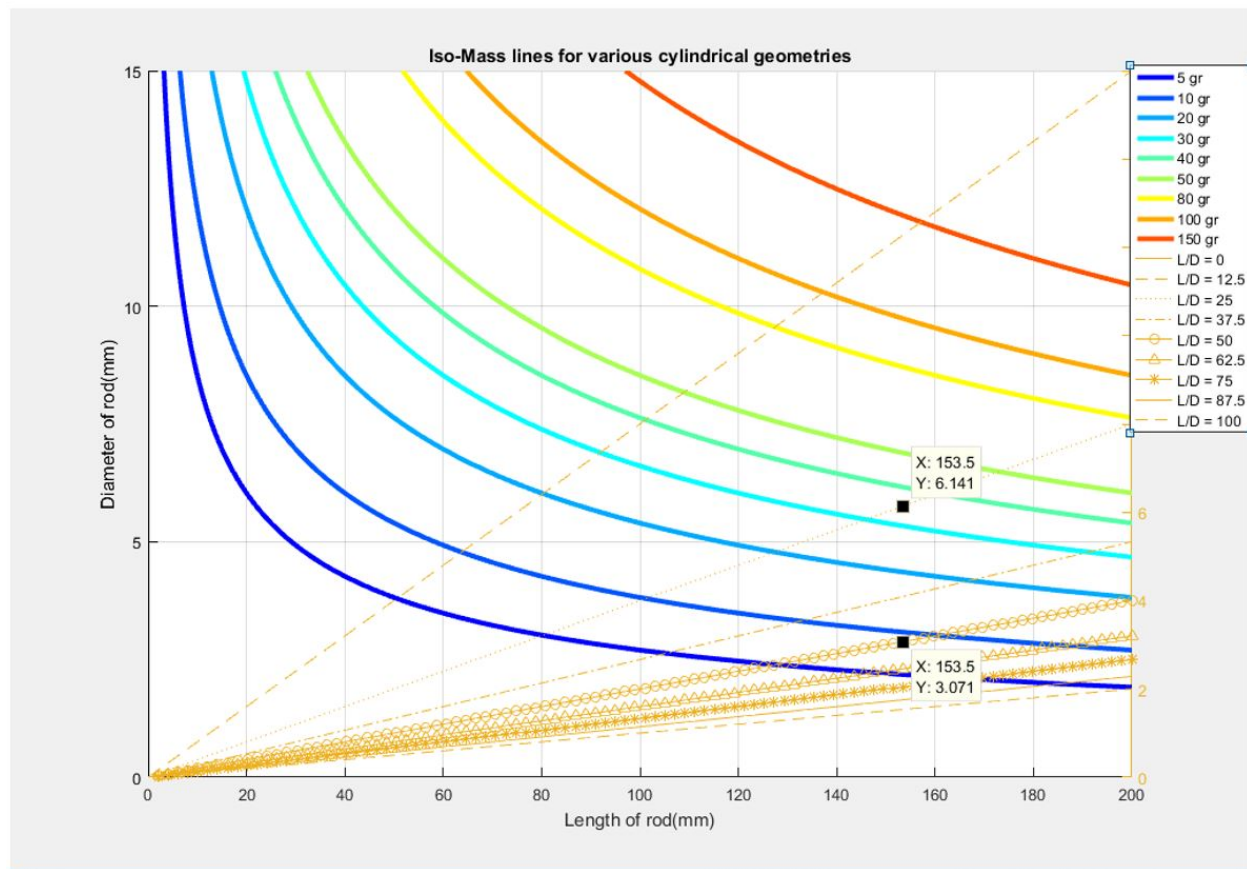


Figure 3.4: Iso-mass and Iso-efficiency lines versus the diameter and length of cylindrical specimens

In order to evaluate the added value of multiple rods versus a single rod, the plot in Figure 3.4 below could be of use and help in the trade study. The horizontal axis of this figure represents the possible length values of a hysteresis rod considered to be installed on the satellite, while the left vertical axis represents the possible diameter values of the same rod. The figure also illustrates two types of lines. Each of the colorful asymptotic lines represent the possible combinations of diameter and length for a rod of a specific weight (and by extension, volume). We call these lines as *iso – masslines*, and are produced by simply considering the computation of a rod’s volume and mass, made out of a typical soft ferromagnetic material, in this case *High Perm 49®*. In the same figure, the straight lines starting from the origin represent what we define as *iso – efficiency* lines. Each one of those lines represents potential combinations of a rod’s diameter and length having the same length-to-diameter ratio (along each line). Now, given the unique constraints of a satellite with respect to mass budget discipline, available volume space, and requirement to detumble the spacecraft using hysteresis dissipative torque, the engineer can better visualize the trade space in this figure and identify potential combinations of length and diameter of rods that will accommodate all above constraints. Using this qualitative tool, the process is meant to

be iterative, as the simulation results that accompany each proffered combination will prove or disprove the compliance to the detumbling requirement.

Based on [26], it is noted that the dipole efficiency (i.e. amount of dipole moment per volume) is approximately proportional to the rod elongation. If we are interested in reducing the mass of the system in order to achieve same hysteresis torque, it is possible to substitute one rod of about 6 mm diameter with two rods of 3.07 mm, and we will have effectively halved the mass. The dipole efficiency per rod will double, while the volume of each rod will shrink to a quarter of its value. Therefore, using two rods will halve the initial mass, for the same performance, while taking note of the additional space that will be required to avoid mutual magnetization. Given the described physical constraint of not being able to use more than three rods at most, coupled with the relatively low mass savings in absolute terms (20 grams instead of 40 grams per axis), the decision to use 1 rod per axis is implemented. In combination with the simulation results provided in Section 3.12, it is proven that this rod schema selected is sufficient to cancel the potential spin-up of a satellite equipped with drag sails.

3.11 Material Selection

Selecting soft ferromagnetic materials is a combination of availability, cost, and performance. The performance within the context of material selection is defined as material's ability to have very high permeability, but at the same time have sufficient hysteresis losses expressed through high flux density. Figure 3.5 shows a material selection matrix based on those criteria, provided by Carpenter, a ferromagnetic material supplier [43].

	Sensitivity (permeability)	Strength (flux density)	Cost
Highest	Nickel-Iron	Iron-Cobalt	Iron-Cobalt
	Silicon Iron	Iron	Nickel-Iron
	Iron	Silicon Iron	Ferritic Stainless
	Iron-Cobalt	Ferritic Stainless	Silicon Iron
Lowest	Ferritic Stainless	Nickel-Iron	Iron

Figure 3.5: Magnetic Alloy Selection Matrix (Carpenter) [43]

Past missions have predominately been using Nickel iron alloys[21], with the main reason being the extremely high permeability in comparison to any other ferromagnetic alloy. Figure 3.6 below from Carpenter Material Selection Methodology illustrates this comparative advantage. Nickel-iron alloys like *HyMu 80*[®] and *High Perm 49*[®] dominate in terms of permeability and are well suited for magnetic hysteresis dissipation purposes.

In addition to the above, specific to the attitude stabilization strategy is the requirement for the material to be both extremely sensitive and "spongy" (high permeability) but also "lossy" in nature and allow for hysteresis losses, contrary to other industries which require minimal losses, such as in power transformers. Some missions have selected *HyMu 80*[®] in the past. *HyMu 80*[®] a nickel-iron alloy with extremely high permeability but relatively low hysteresis losses. TRANSIT-1B and TRANSIT-2A used *AEM-4750*, also known as *High Perm 49*[®]. A nickel-iron alloy, the latter is more lossy than *HyMu 80*[®] with a larger hysteresis major loop, yet with lower maximum permeability. Generally speaking, the larger the area of a material's H-B major loop, the larger the energy losses will be during the material excitation. The exact trade-off is difficult to make without simulating the exact conditions of excitation because the magnetic field strength of the Earth is far lower than the magnetic field used to characterize the outer major loop of the material. This becomes clear after the experimental characterization of the SFL rods, and the produced H-B plot in Section 3.13. Thus, choosing materials that maximize hysteresis losses (i.e the area enclosed by the major hysteresis loop) given a magnetic field excitation is very important. Recently, [28] demonstrates that Grain Oriented (GO) silicon-iron (*SiFe*) alloys show much promise in maximizing energy losses for attitude detumbling. However, the material has no flight heritage in any space mission yet.

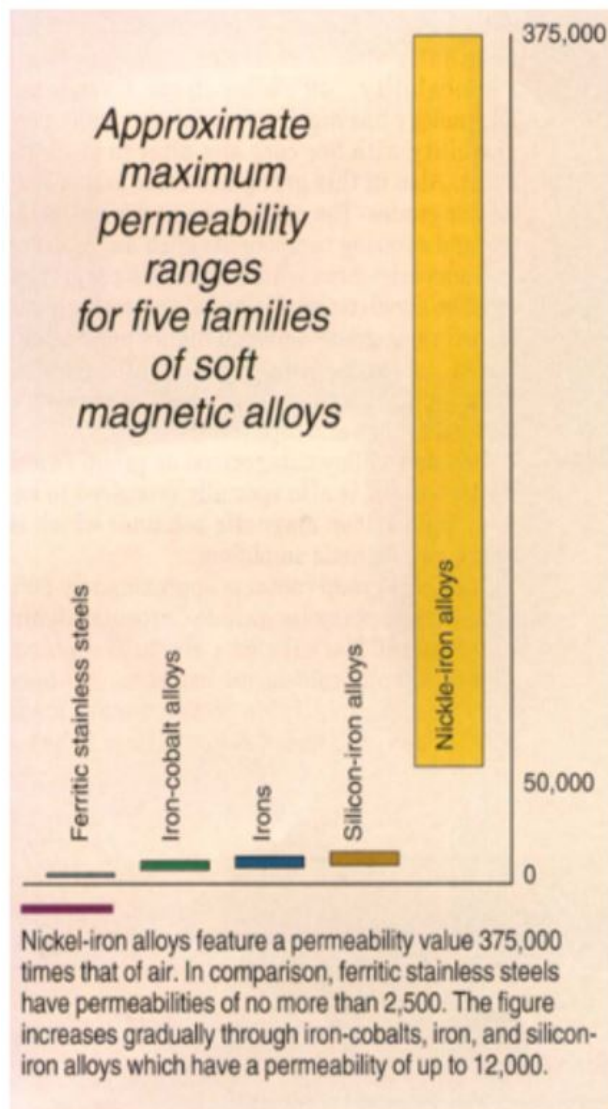


Figure 3.6: Comparison of permeability for different alloys. The vertical axis represents how many times the permeability of vacuum each material family demonstrates. [43]

HyMu 80[®] rods have been characterized for attitude detumbling of CubeSats by several sources, specifically by [26], [27] and [30]. The experimental setup of the latter reference shows more promise in validity of measurements due to the experiment being conducted in a Helmholtz cage that can isolate the measurements from the Earth's magnetic field. *High Perm 49[®]* rods will also be considered for the simulations, since it is the material of the rods in the laboratory's current inventory and due to its successful flight heritage.

3.12 Simulation Results

Table 3.1 below illustrates the important initial conditions of the simulations performed. Simulation #1 is the baseline simulation that was shown at the end of Chapter 2, yielding the largest spin-up observed by the spin-up permutation analysis (Section 2.6). All subsequent simulations use the same initial conditions for satellite body rate, attitude, and other modelling effects, with the only differences among them being the three parameters that characterize the hysteresis rods installed. The exact hysteresis parameters used for each simulation are shown in Table 3.1.

Table 3.1: Simulations performed and their initial conditions. Angular velocities are about the Sun-pointing axes

Sim Index	Rods per satellite body axis	Material	h_c	b_r	b_s	Motivation
1	0	N/A	N/A	N/A	N/A	Baseline sim, with spin-up conditions
2	1	<i>HyMu 80</i> [®]	0.3381	0.0006	0.300	Experimental properties [30]
3	1	<i>HyMu 80</i> [®]	12.000	0.004	0.025	Experimental properties [26]
4	1	<i>HyMu 80</i> [®]	1.590	0.350	0.730	Theoretical properties [30]
5	1	<i>High Perm 49</i> [®]	2.860	0.961	1.490	Theoretical properties
6	1	<i>HyMu 80</i> [®]	0.3381	0.004	0.300	Based on Sim #2, with larger b_r
7	1	<i>High Perm 49</i> [®]	0.609	0.0034	0.612	Scaled, based on Sim #2 and #6

As discussed, the simulations are performed using the NorSat-3 mechanical structure properties, with the setup as shown in the previous chapter in Figures 2.7 and 2.8. With the exception of Simulation #1, all simulations assume one hysteresis rod per spacecraft axis, of length 152.3 mm and diameter 6.4mm, which yield a length-to-diameter ratio of approximately 24, as shown in Equation 3.27. The mass of each rod is approximately 40 grams.

By observation of Table 3.1, the theoretical magnetic parameters for *HyMu 80*[®] alloy vary substantially with the measured results from [30] or [26]. Treating [30] as a more confident measurement due to the number of samples tested and the magnetic isolation of the setup in the Helmholtz coil, large discrepancies were still observed by the authors in

magnetic properties even within the same lot of rods. However, it is important to note that such discrepancies should be expected in part because the theoretical magnetic parameters are a result of subjecting the specimen to maximum field amplitudes, in the order of 800 A/m . The averaged measured properties were collected from field cycles of 100 A/m , so it is expected for all measured values to be smaller because the specimens are cycled over a hysteresis loop that resides completely within the theoretical major loop. Discrepancies between specimens and the particularly high deviation of remanence b_r from the theoretical values have triggered the need for an experimental setup similar to [30] in our case, since rod performance needs to be validated at on-orbit magnetic field levels, in the order of 20-40 A/m . The test methodology followed in Section 3.13 relies largely on the same principles as that of [30].

Simulation #2 uses coercivity, remanence, and saturation values as those were determined through experimental fitted measurements from [30] for *HyMu 80[®]*. Simulation #3 uses another set of experimental values for *HyMu 80[®]*, as reported by [26]. Simulation #4 uses the theoretical values of *HyMu 80[®]* as reported by [30]. Simulation #5 uses the theoretical values of *High Perm 49[®]* as reported by the manufacturer of the SFL cylindrical rods in the inventory. Simulation #6 uses the same coercivity and saturation values as Simulation #2, but with a slightly higher remanence, to assess the effect of that parameter on the dissipative action in the simulation. This remanence value is selected at the upper range of experimentally observed values from the same source. It should be noted that *HyMu 80[®]* very high permeability, but very low coercivity and very low remanence, which minimizes losses so much as to not make it a good damping material, regardless of it having been used in some spacecraft missions. Parameters of Simulation #7 were determined using the following methodology. Based on the reported discrepancy of [30] between the theoretical and experimental values of b_r , b_s and h_c for *HyMu 80[®]*, a down-scale factor was determined for each of these parameters between their higher theoretical values and their lower experimentally determined value. Then, these down-scale factors were applied on the theoretical properties of *High Perm 49[®]*, in order to produce a set of pseudo-experimental magnetic properties, under the assumption that *HyMu 80[®]* and *High Perm 49[®]* would share the same discrepancy, if tested with the same methodology. This is done as an effort to introduce the same amount of margin to another material, for which no experimental data was found. After experimental testing of SFL's rods in Section 3.13, it is proven that the values match closely the results of the scaled-down remanence and coercivity, while saturation does not seem to scale linearly, but is well within acceptable margins. In the end, the experimental values of the rods prove to be adequate in dissipating the spin-up behavior.

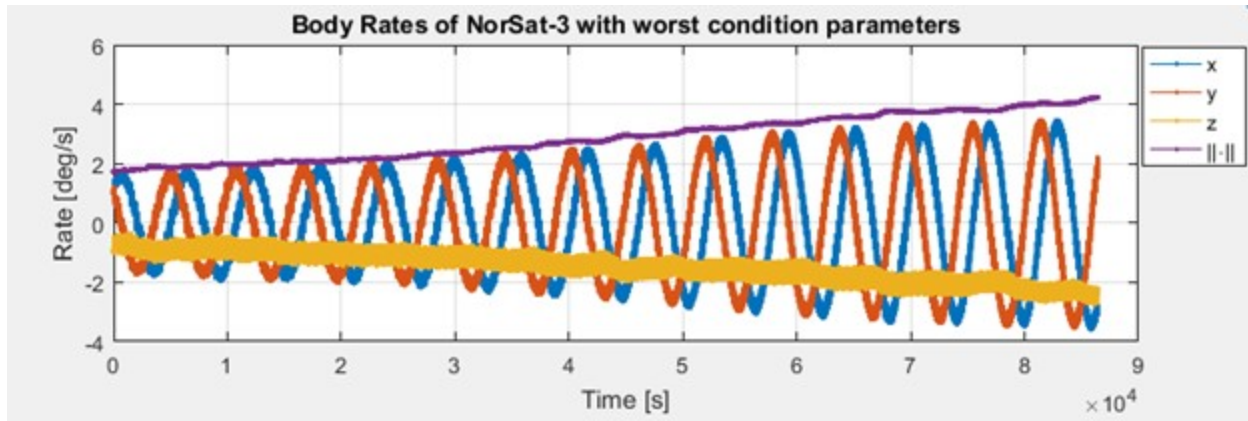
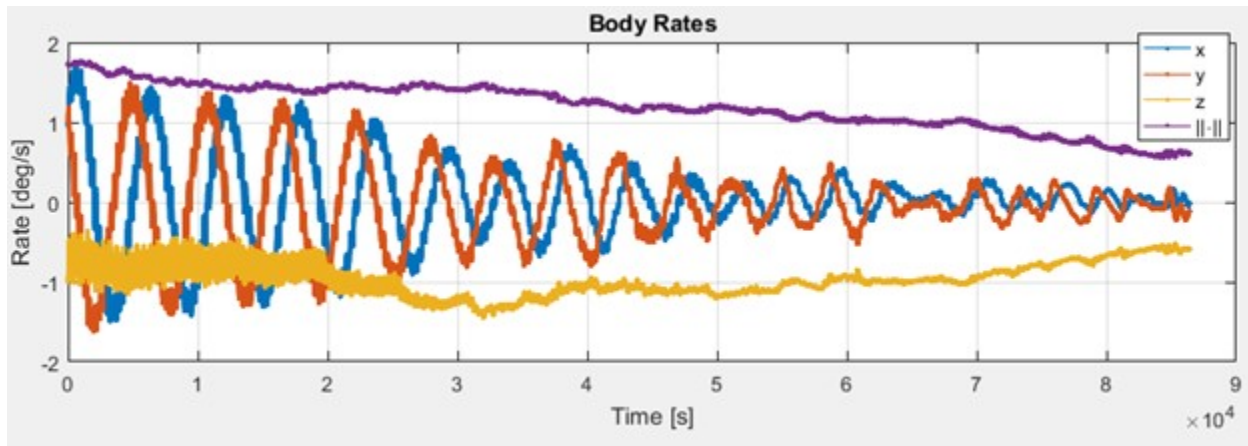
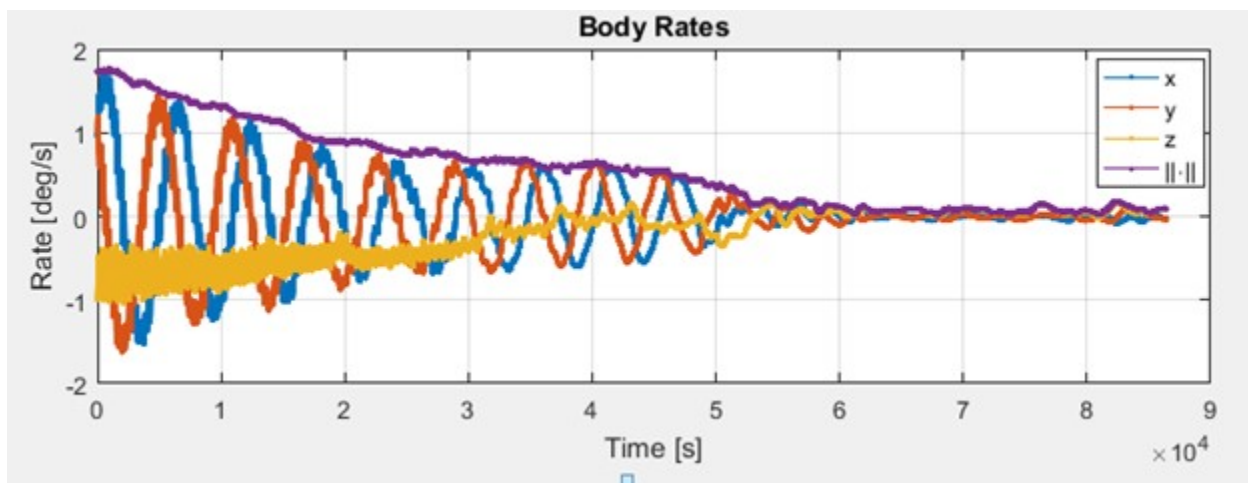


Figure 3.7: Simulation #1 Body Rates, no rods installed

Figure 3.8: Simulation #6 body rates, using *HyMu 80*[®] rodsFigure 3.9: Simulation #7 body rates, using *High Perm 49*[®] rods

The results of Simulations #1, #6, and #7 are shown in Figures 3.7, 3.8, and 3.9 above.

Each figure shows the satellites body rates with respect to $\mathcal{F}_i$, expressed in $\mathcal{F}_b$, as well as the angular velocity norm, shown in purple color. In general, *High Perm 49[®]* seems to perform better than *HyMu 80[®]* mainly due to the overall higher coercivity and remanence, under the assumption that the same scale-down factors that apply to *HyMu 80[®]* will also apply to *High Perm 49[®]*. *High Perm 49[®]* is the same material used in SFL’s NTS mission, designed around the GNB bus.

The simulation is performed using a simulator based on the core functions of the Mirage simulation environment, but with the added functions for computing the hysteresis damping torque, as described in 3.9.2. All environmental disturbances are being considered, and the worst-case initial conditions that trigger the spin-up are used, as explained in Chapter 2.11, Section 2.6. The next section expands on the experimental verification of the coercivity, remanence and saturation values used for *High Perm 49[®]* alloy.

3.13 Experimental Validation

This section illustrates the test results produced by in-house testing of the hysteresis rods. Six hysteresis rods of *High Perm 49[®]* alloy were tested. The material is also known as AER 4750, or MIL-N-14411C. Each rod has a diameter of 6.4 mm, length of 152 mm and weighs approximately 40 grams.

Few literature resources within the space community exist for the experimental setup required for hysteresis rod testing. A process using a single solenoid and a magnetometer attached at its one end is described in [26]. A similar setup is also used in [29]. ASTM standard ASTM A773/A773M provides a test method for direct-current magnetic property measurements of low-coercivity magnetic materials using hysteresis graphs [44]. However, the A773 standard suggests that the set-up is not optimal for hysteresis testing of nickel-iron alloys due to errors associated with integrator drift. ASTM suggests the use of ASTM A596/A596M to test soft ferromagnetic materials using the ballistic method, even though an oscilloscope with data capture and math function capabilities could avoid the integrator drift problem [45]. The procedures of the latter standard were used to perform the tests.

The procedure involves the electromagnetic induction of a secondary coil, by driving a sinusoidal current through a primary coil. The hysteresis rod fits tight within the secondary coil. The setup is placed within a Helmholtz cage that has been previously tuned with the help of a high-fidelity calibrated magnetometer to cancel out the local magnetic field within its bounding volume. The whole process is performed at SFL’s magnetically clean chamber, in order to minimize the introduction of parasitic noise effects in the test. The magnetically clean chamber is part of the SFL ground facilities. It is a room shielded with in-wall steel beams, where magnetic measurements can be performed, isolated from the rest of the laboratory where spurious electromagnetic noise exists. The chamber also serves

as a "dark room" where solar cells as well as other attitude sensors are tested. Within the chamber, a Helmholtz coil exists, driven by three independent power supplies. The Helmholtz coil is used to calibrate sensitive attitude determination sensors, such as magnetometers. The Helmholtz coil has been used as part of the test, to ensure a magnetically clean environment during the hysteresis rod characterization. As part of the experimental setup an operational amplifier is used at the output signal of the secondary coil to condition the signal. Both input and output signals are captured by an oscilloscope with integrator function capability and the results were stored in an external media for processing. Gerhardt and Palo [30] provide a detailed description of the experimental setup used, showing some resemblance to ASTM 596 practices. They used a Helmholtz cage as well as an instrumentation amplifier with the secondary coil to capture the B-H data through an oscilloscope.

A block diagram of the in-house experimental setup is shown in Figure 3.10 below. The block diagram features three types of connections, annotated with three different colors. Starting from the top-left, a computer drives three independent programmable power supplies, through the purple data connection $S_PC.1$. The power supplies are driven in such a way so that to maintain a stable sinusoidal magnetic field within the volume of the Helmholtz cage, by using the calibrated magnetometer in a closed-loop. The data connection with the magnetometer measurements to the PC is annotated as the purple line $S_PC.2$. Each of the programmable power supplies provides power to each coil in the 3-axis Helmholtz coil assembly, through the red power connection annotated as $P_HC.1$. The hysteresis rod with the sense coil is mounted inside the volume of the Helmholtz cage, and aligned with one of the coils axes. The applied voltage on this driving coil is input to the first channel of the oscilloscope, and this voltage signal connection is annotated as the green line V_PS . The secondary coil that surrounds the hysteresis rod is made out of AWG-40 copper-wire. It senses the response of the hysteresis rod through the externally applied field from the Helmholtz coil, by producing a voltage across its ends. This low voltage of the sense coil is provided as input to the low noise amplifier through the signal connection V_SC to perform signal conditioning. In the end, the voltage signal from the output of the low noise amplifier is input to the second channel of the oscilloscope, through the connection labeled as V_AMP .

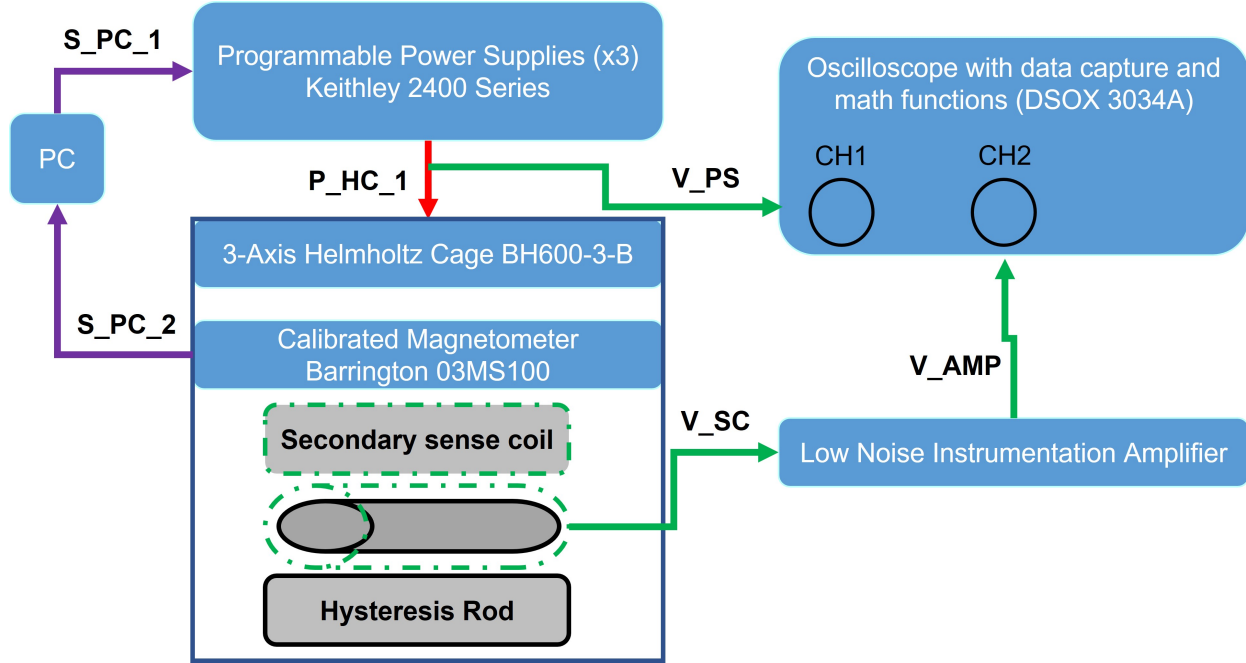


Figure 3.10: Block Diagram of the suggested experimental setup for hysteresis rod testing

It is noted that the experimental setup makes use of the Helmholtz cage for two reasons: First, it provides isolation from the external magnetic field environment by nullifying it, with the use of the 3-axis magnetometer as a feedback sensor. In addition, it provides an accurate source of a sinusoidal magnetic field within the cage's volume, by programmatically controlling the power supplies, operating in current source mode. In the middle of the cage, the secondary sense coil is installed such that its axial direction is parallel to one of the coil's axis as well as to that of the magnetometer's axis. The reason for this mounting solution is because it will require only one power supply to provide the required switching current/magnetic field, about a single axis. The hysteresis rod under test is placed within the sense coil, prior to its installation in the cage. With respect to the low noise amplifier, the AD8421 Op Amp Evaluation Board was used, due to its superior signal conditioning characteristics. The integration of the signal from the low noise amplifier (connection V_{AMP}) is performed using the oscilloscope. The signal needs to be integrated with respect to time before further being processed to a value representing magnetic flux density. Consider the secondary sense coil above, comprised of n number of turns, with a circular cross sectional area a . In addition, we define e as the induced voltage sensed by the secondary coil, amplified by the LNA (low noise amplifier), in units of Volts db . Then, the magnetic flux density difference Δb between current timestep t_1 and the previous timestep t_0 is given by

the relation:

$$\Delta b = -\frac{1}{na} \int_{t_0}^{t_1} e dt \quad (3.28)$$

In order to ensure that the measurement will be characteristic of the hysteretic behavior of the material, the switching frequency of the magnetic field needs to be sufficiently low. Gerhardt and Palo, after discussing the test method used by Ivanov conclude that a frequency of 1Hz is sufficient, as it represents the golden mean between acquiring accurate measurements and the ability of the oscilloscope to sample both signals with adequate resolution. The latter is also known as ADC quantization, and represents the sensitivity of the oscilloscope in capturing faint signals with acceptable resolution. In addition, the switching frequency is related to the spacecraft's frequency of rotation, which should not be more than 0.1 Hz in magnitude. Therefore, switching frequencies in the order of 0.1-1 Hz should be implemented, with a preference to lower frequencies. The signal generator that was used was capable of frequencies as low as 20 mHz , which is considered as a quasi-static. The frequency of the applied sinusoidal signal was 28 mHz . Figure 3.11 below illustrates the signal generator that was used. The signal amplitude was 40 A/m resembling approximately the maximum magnitude of the magnetic field of the Earth in a Low Earth Orbit.



Figure 3.11: Tenma 72-7650 signal generator

Due to potential residual magnetization within the rod, the experimental procedure involved a degaussing process prior to the main test. Degaussing eliminates any direct current offset in the induced sensing voltage that would lead to the generation of asymmetric, not-centered hysteresis loops in the B-H plot. Hysteresis rods can maintain their magnetization to some extent depending on their short and long term storage environment, their distance from other ferromagnetic materials, handling, and any prior testing that has been performed.

Degaussing is a suggested technique that is also included in the ASTM A773 and ASTM A596 standards. For more information about the two standards, consult [44] and [45].

The process of degaussing involves the sourcing of a sinusoidal magnetic field with an exponentially decaying amplitude over time, starting from an amplitude that is large enough to ensure that little residual magnetization will remain in the end. Degaussing attempts to erase the hysteretic memory of the material as much as possible, so that the measurement that follows can start in the middle of the B-H plot and not outside of the expected hysteresis loop at the cycle amplitude of choice. Figure 3.12 demonstrates a degaussing profile. Figure 3.13 shows the sense coil assembly that was used during the experimental procedure. This assembly was installed within the Helmholtz cage and aligned with the coil that produced the sinusoidal magnetic field.

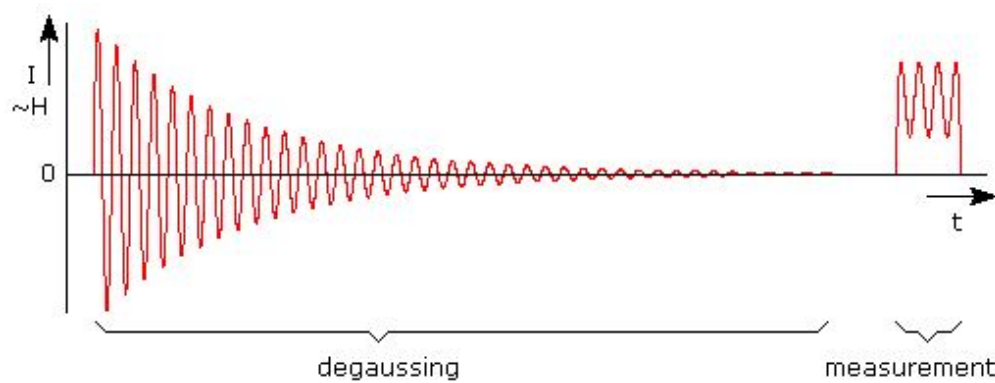


Figure 3.12: Degaussing profile. Prior to testing the rod, each sample needs to be degassed so that the any residual magnetism is completely removed from the specimen. [46]

The experimental procedure that has been followed is a robust method to characterize hysteresis rods in an isolated environment, at low frequencies which can be programmatically controlled to change sinusoidal or even match exact on-orbit changes in the magnetic field. Eddy current losses will be minimized due to low switching frequencies and the use of the instrumentation amplifier allows for testing at very low magnetic field strengths, provided that the common mode noise is sufficiently low.

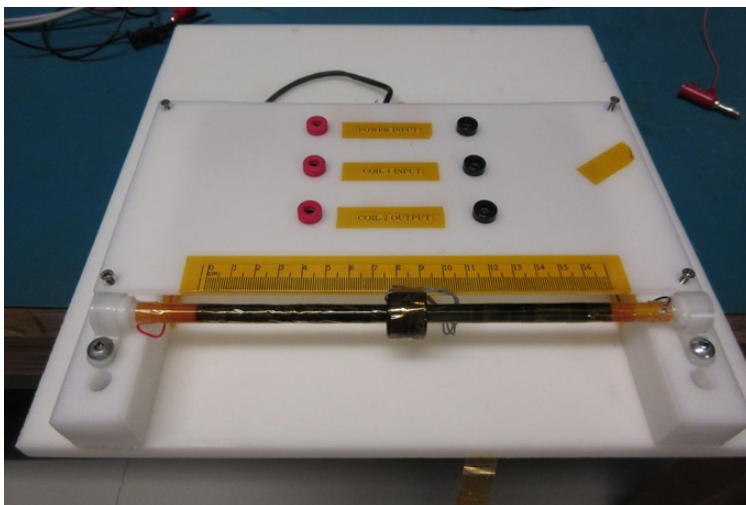
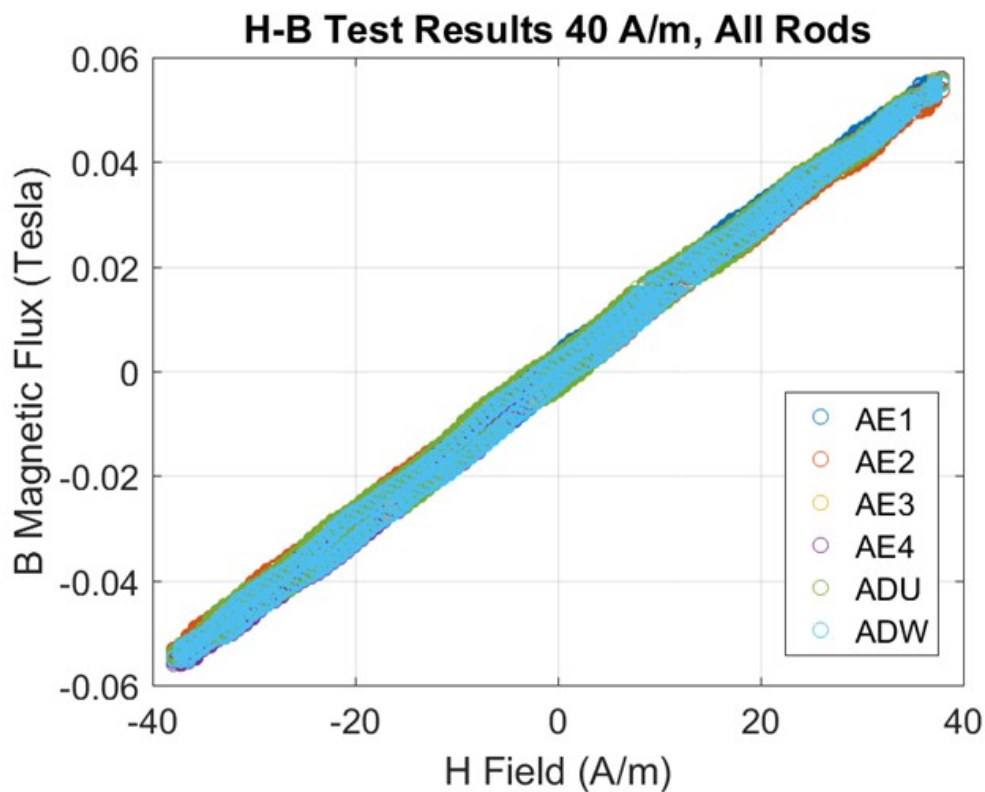


Figure 3.13: Primary and Secondary coil setup

The results of the hysteresis rod testing are processed to produce the H-B plots, presented in Figure 3.14 below. The H-B plot is also known as a *hysteresis graph*. The horizontal axis represents the value of the sinusoidal externally applied magnetic field, which is the independent variable. The vertical axis illustrates the value of the induced magnetic flux density within the hysteresis rod, as a response to the externally applied field.

Figure 3.14: *High Perm 49[®]* hysteresis rod testing results

The plot illustrates the results of six cylindrical *High Perm 49[®]* hysteresis rods. The legend that accompanies the plot illustrates the part number that each hysteresis rod corresponds to. At SFL, all hardware follows a naming convention for parts, facilitating unit-testing and inventory purposes. Parts are typically named after a three-character alphanumeric code.

By observing the results, we see that all rods produce consistently similar curves. This proves that the cylindrical specimens have similar material properties. In addition, it is noted that the produced H-B curves are centered about the axis origin. This proves that the rods were effectively degaussed at the beginning of the test procedure. It also proves that the use of the Helmholtz cage was successful in isolating the hysteresis rod and sense coil from the Earth’s magnetic field, while applying the sinusoidal field at the same.

By further examination of Figure 3.14, we notice that the curve of each rod does not exactly look like an arctangent function. This is expected, since according to [23], *the arctangent model approximates the characteristic major loop* of the material. When the material experiences magnetic fields that are smaller than the high magnetic fields that were used to characterize the rod, the material’s response will reside within the major loop, forming leaf-shaped loops that slowly reassemble to an arctangent function as the magnetizing field increases sufficiently [23]. Furthermore, the low excitation magnetic field used produces curves that reside well within the characteristic sigmoid major loop shown in Figure 3.1. With regards to the magnetic parameters, we note that the remanence b_r for all rods is observed to be approximately 0.004 Tesla, the coercivity h_c is approximately 2 A/m, while the saturation b_s is approximately 0.056 Tesla. With the exception of saturation b_s , those values are all higher than the values used for *High Perm 49[®]* in Simulation #7, which represent a very conservative estimation of the rods dissipating capabilities. More specifically, we observe that, in relation to Simulation #7, the H-B curve that we have experimentally determined is wider (due to higher h_c) and also with slightly higher remanence, but with a lower saturation by a factor of 10. The discrepancy of the saturation means that the “pseudo-scaling” that was applied from the *HyMu 80[®]* material was not linear in nature. Regardless, it is noted that with regards to an H-B plot, the bigger the area that a hysteresis loop occupies, the larger the energy losses are. With a higher coercivity and remanence but smaller saturation, there is no quick conclusion that can be drawn at this point about whether the SFL rods are better or worse in dissipating rotational energy. Therefore, an additional simulation is needed with the aforementioned experimentally determined values of Figure 3.14.

The spin-up simulation was performed one last time with the averaged experimentally determined values of coercivity, remanence and saturation to observe the effect of the exact rod parameters on the detumbling performance. The results are presented in Figure 3.15 below, showing the satellites body rates with respect to $\mathcal{F}_i$, expressed in $\mathcal{F}_b$, in addition to the angular velocity norm, in purple color. The simulation was performed for a longer

time span of ten days, and not just for one day, which was the time span of the previous simulations in Table 3.1.

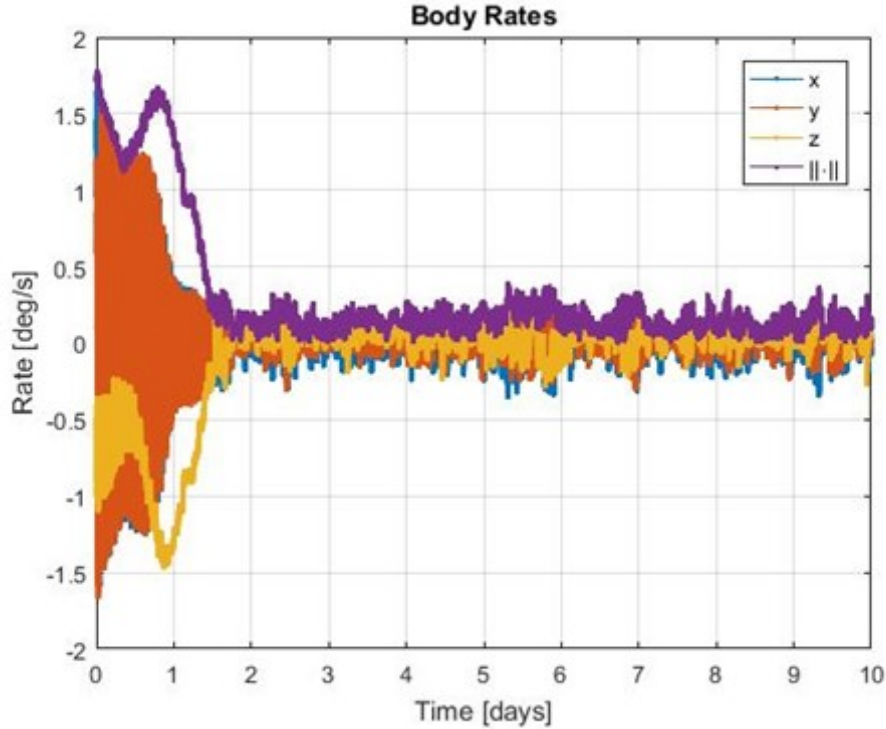


Figure 3.15: Final Simulation of spacecraft detumbling using averaged experimental hysteresis rod parameters

The addition of the rods with the experimentally determined values leads to complete stabilization of the body rates to within 0.5 deg/s per axis, with a steady state angular velocity magnitude no larger than 0.5 deg/s in total. This proves that the rod diameter, length, and material all contribute to a set of coercivity, saturation and remanence such that a spin-up initial condition of approximately 1 deg/sec per day can be avoided exclusively with the use of passive means. At this point, it is important to recall that this simulated spin-up and cancellation observed is based on careful setup of the initial conditions of initial body rates, attitude, and potential drag sail twist by a few degrees. Effectively, during the simulation, the forcing condition is still constantly present, as seen by the spin-up results of Simulation #1, mainly due to the assumption that the twist of the sails on the satellite structure is permanent, and unchanging. If other factors end up applying a constant torque on the satellite such that it spins up at a rate that is higher than simulated, then the stabilization of the satellite will take a longer time, and it is possible that more hysteresis rods will need to be added to ensure an adequate amount of dissipative torque. This will depend on the amount of drag sails installed, satellite geometry, and future improvements on the rigid body or integration of flexible dynamics within the simulation model, for better accuracy.

If we compare the results of Simulation #7 (Figure 3.9) with the results of this simulation on Figure 3.15, it is clear that in the latter case the stabilization occurs slower. In Simulation #7, stabilization was achieved within less than a day, while in the latest simulation, stabilization was achieved in approximately 1.5 days. Recall the discussion on the discrepancy between experimental magnetic parameters of our setup, and the parameters of Simulation #7, earlier in this section. Even though the coercivity and remanence of our rods were higher, the saturation of our setup was also determined to be a factor of 10 lower than Simulation #7. Also recall that the process of determining the parameters of Simulation #7 was through a judicious scale-down factor based on theoretical and experimental values of another material, due to lack of experimental data for *High Perm 49[®]* prior to testing our rods. This proves that our experimentally determined values are very conservative, with a carefully designed characterization procedure, and therefore more realistic. In fact, the literature review on hysteresis parameter characterization suffers from overestimating the dissipative capabilities of hysteresis rods. This has been briefly explained in the beginning of this chapter. Recall that [27] and [32] overestimated the stabilization time using hysteresis rods by as much as 60 days, which was confirmed from on-orbit telemetry afterwards. The experimentally determined magnetic parameters of the SFL rods allow for a more realistic approach on the dissipative capabilities of hysteresis rods, thus providing much clarity of the trade space of options. Corrective measures and improvements may include the addition of more rods, maintaining stricter mass discipline, changing drag sail configuration, or exploring other means of passive stabilization.

Finally, the hysteresis parameters and the rod geometry of the latter simulation were input to the Mirage simulations to assess the impact of the rods in the overall performance of the NorSat-3 active attitude control system. The simulations illustrated that all performance metrics were not significantly altered and all attitude control requirements were still met. This is expected, since the introduced torque due *High Perm 49[®]* is purely dissipative in nature, and would not contribute to a disturbance that would require feed-forward control or other means of active control to counter.

3.14 The Hyperbolic Model for Major Hysteresis Loops

As previously mentioned, the arctangent model used to perform the simulations of Section 3.12 is used as the basis for all simulations performed in [23], [25], [27] and [33]. In order to fit the experimental data to such a model, the user has the ability to vary the three magnetic parameters of coercivity, saturation and remanence. In the case where the experimental data may not be able to be fitted adequately, literature suggests that another simple model can be used instead, known as the hyperbolic model. It is noted that this model is provided for completeness of literature review, and also for future work on validation of the hysteresis rod

testing results, as higher fidelity input to the simulations.

The hyperbolic model approach is based under the simple assumption that the magnetization process of a soft ferromagnetic material is the combined result of three distinct magnetization processes that occur in parallel and are each associated with a different magnetization mechanism of the domain walls; those processes are the domain wall movement, the domain rotation and the domain wall annihilation and nucleation. Each of these processes dominates the low, middle and near saturation region of magnetization respectively, and they can be separately modeled as hyperbolic functions. This model has been successfully used in a number of applications with well documented sources in the literature [47], [48] [22]. Among the space industry literature, the model has not yet drawn much attention.

The use of the hyperbolic tangent function to describe the hysteresis phenomena is not randomly selected. In fact, a formulation of the hyperbolic tangent function is an approximate solution to the integral form of the Preisach elementary operator, also known as Everett's integral. The reader is directed to [22] for more information on how the function is derived. In summary, the hyperbolic model is a simplified yet robust approximation to the stochastic Preisach model, considered as the most accurate model that exists to describe hysteresis behavior. The Preisach model takes into account the complete past magnetization history of the material to predict the magnetization, whereas the simpler models avoid this approach, to sacrifice accuracy for computational load, by considering only the immediately previous state of the material's induced flux density, and not its complete history.

3.15 Conclusion

In this chapter, we have demonstrated that the installation of three orthogonally mounted hysteresis rods within a NEMO class satellite equipped with two SFL drag sail modules can eliminate an undesirable spin-up condition that stems from an unfavourable initial condition of body rates, attitude, and potential drag sail twist of by a few degrees with respect to the satellite structure. After a thorough literature review on the phenomenology and fundamentals of magnetic hysteresis behavior of soft ferromagnetic materials, we have formulated an equation for computing the total dipole moment of a satellite due to the presence of hysteresis rods of arbitrary number, shape, and orientation within a satellite's structure. The equation has been used to model the effect of the hysteresis rods on providing dissipative rate-dependent torque, using theoretical as well as experimentally determined magnetic parameter values of remanence, coercivity and saturation. The rods considered in the experiment and simulation are 152mm length, 6.4mm diameter and made of *High Perm 49[®]*, which is a soft ferromagnetic material. Rod dimensions adequately fit within the mechanical structure of a NEMO class spacecraft bus. by comparing simulations using *HyMu 80[®]* and *High Perm 49[®]*, it is shown that *High Perm 49[®]* demonstrates better dissipative charac-

teristics than *HyMu 80*[®], mainly due to the higher coercivity and remanence. The latter characteristics assist in the increase of energy losses and the dissipation of rotational energy into heat. In addition, an experimental procedure for characterizing hysteresis rods in current SFL inventory has been performed. The verification process included two important features, namely the use of a Helmholtz case, and the use of a very low frequency sinusoidal signal with same amplitude as that experienced in Earth orbit. The results were fed back to the simulation to validate the stabilization performance and demonstrate compliance to the detumbling requirement.